The following rules must be fulfilled by the type-checking phase.

Characters are subtypes of integers, and integers are subtypes of doubles. The subtyping relationship is transitive (e.g., characters are subtypes of doubles) and reflexive (i.e., any type is a subtype of itself).

For expressions:

- The +, -, * and / binary operators can be applied to characters, integers and doubles. If either operand is a double, the result is a double; otherwise, it is an integer.
- The % operator can only be applied to characters and integers. An int is returned.
- Logical operators (&&, || and !) can only be applied to expressions promoting to (subtype of) integer, and return an int.
- Comparison operators (>, <, >=, <=, == and !=) are applicable to characters, integers, and doubles; they always return int.
- Unary minus can be applied to char and int (returning int), and double (returning double).
- The cast operator can only be applied to built-in types (excluding void).
- The [ ] operator requires the first operand to be an array and the second one to be promotable to integer.
- The . operator can only be applied to records (structs). The name of the field must be defined in the record definition. The returned type is the field type.
- For function invocation, the number of arguments must be the same as the number of parameters. The type of each argument must be a subtype of the type of the corresponding parameter. Note that parameter and return types must be built-in (void is only valid for the return type).
- Any other combination constitutes a semantic error.

For statements:

- An assignment is valid if the type of the right-hand side expression is as subtype of the type of the left-hand side expression. Both types must be built-in types (excluding void) and the left-hand side expression must be an l-value.
- Only built-in types can be read and written. Expressions used in read operations must be l-values.
- The conditions in while and if statements must be int.
- The type of the returned expression must be a subtype of the return type declared in the enclosing function.